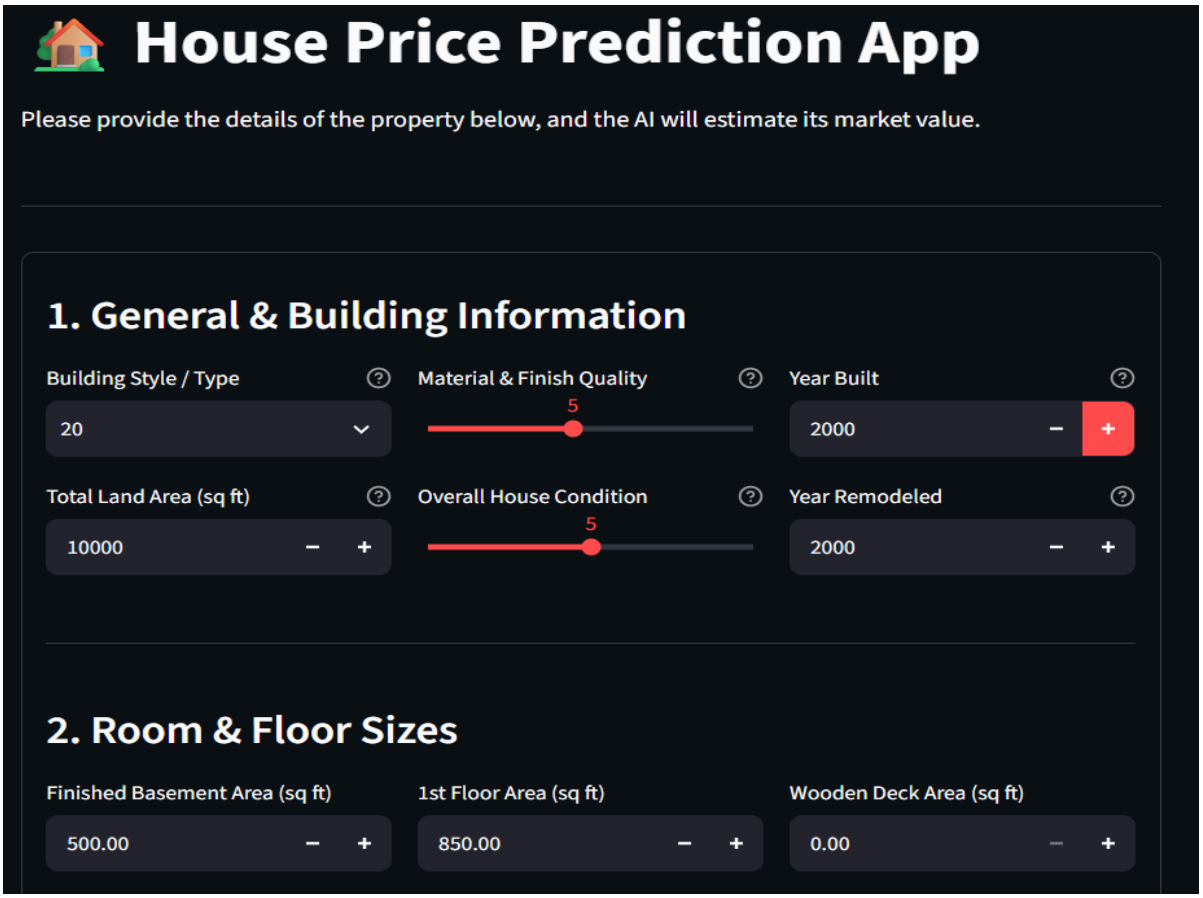
House Price Predictor

A Professional Machine Learning Solution for Real Estate Valuation

This repository contains a comprehensive House Price Prediction system, utilizing advanced regression techniques and a streamlined user interface to provide accurate property valuations.

🖥️ Streamlit Web User Interface

The application features a clean, responsive web interface built using Streamlit. It organizes inputs into logical sections—General Information, Room & Floor Sizes, Rooms/Bathrooms/Garage, and Additional Features—with helpful tooltips and real-time metric updates.



📊 Dataset Information

- **Source Dataset:-** https://www.kaggle.com/datasets/jenilhareshbhaighori/house-price-prediction/data?select=house_cleaned.csv
- **Target Variable (y):** `SalePrice` (Final selling price of the house).
- **Feature Engineering & Preprocessing:**
    - **Identifier Removal:** Dropped the unique identifier column (`Id`) as it adds no predictive value.
    - **Data Integrity:** Checked for missing values, handled data types, and split the dataset using an 80-20 train-test split (`random_state=42`).
    - **Normalization:** Utilized `StandardScaler` where necessary to normalize feature scales (mean = 0, standard deviation = 1) for distance-sensitive algorithms.

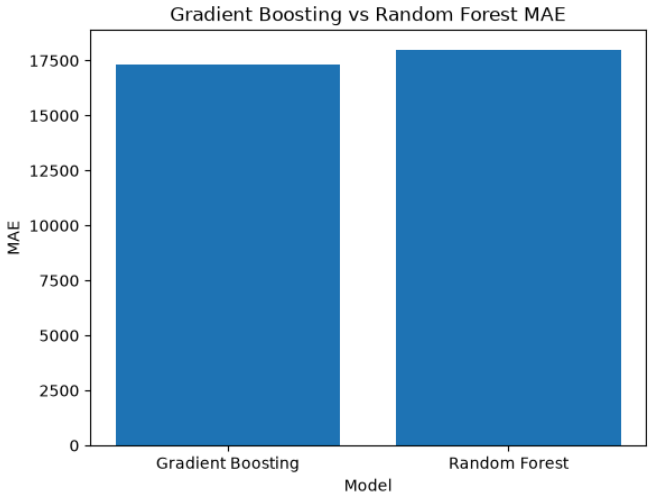
🧠 Machine Learning Model & Algorithm Selection

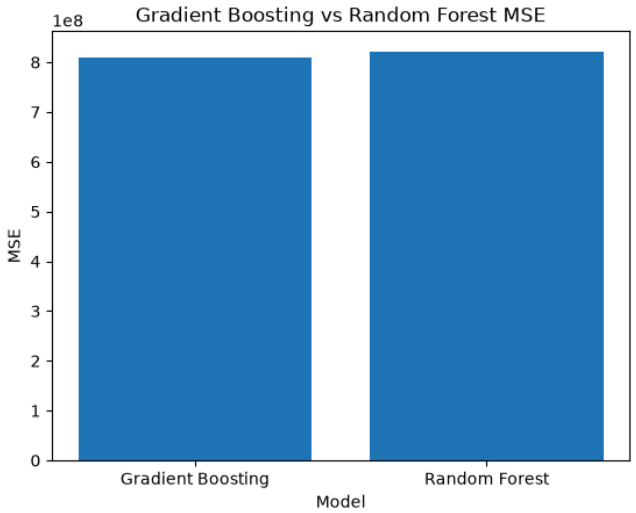
To achieve optimal predictive accuracy, three distinct machine learning regression models were trained, evaluated, and compared:

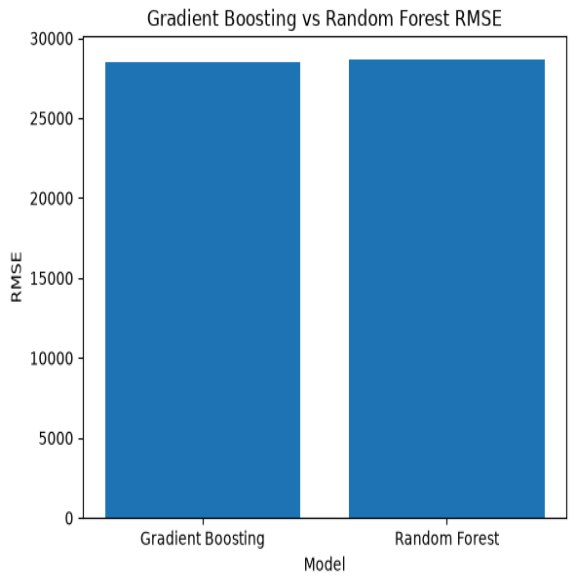
Model	Technique	Key Characteristics
Linear Regression	Baseline	Understands fundamental linear relationships between features.
Random Forest	Bagging Ensemble	Uses multiple decision trees (n_estimators=300) to capture non-linear patterns.
Gradient Boosting	Boosting Ensemble	**Selected Production Model.** Sequentially corrects errors from previous trees.

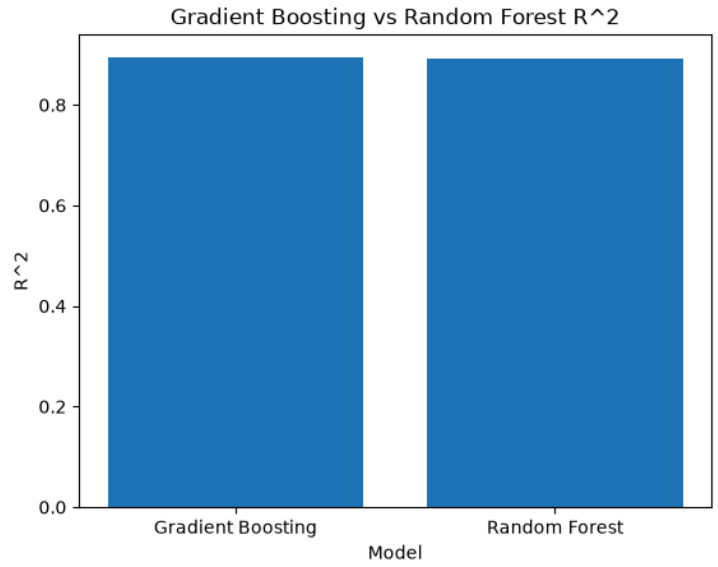
Why Gradient Boosting?

- **Sequential Error Correction:** It builds trees sequentially, where each new tree specifically targets and minimizes the residual errors of the previous trees.
- **Superior Performance on Tabular Data:** Real estate pricing involves complex, non-linear interactions (e.g., how age, square footage, and overall quality compound together). Gradient Boosting models excel at capturing these nuances without requiring heavy manual feature scaling.
- **Hyperparameters:** `n_estimators=300`, `learning_rate=0.05`, `max_depth=3`, `random_state=42`.









📈 Evaluation Metrics & Model Comparison

Models were evaluated using standard regression metrics to ensure reliability:

- **MAE (Mean Absolute Error):** Measures the average magnitude of absolute prediction errors.
- **MSE (Mean Squared Error):** Squares errors, heavily penalizing larger outlier mistakes.
- **RMSE (Root Mean Squared Error):** Interpretable error metric in the exact same units as SalePrice ($).

- **R² Score (Coefficient of Determination):** Represents the proportion of variance in the target variable explained by the features (closer to 1.0 is better).

  All nurmiacal data:-
- R2: 0.894401570332499
- MAE : 17295.32
- MSE : 809974402.52
- RMSE : 28460.05
- R2 : 0.89

Gradient Boosting consistently yielded the highest R² score and lowest error margins compared to baseline and standard random forest configurations.

🛠️ Tech Stack & Libraries

- **Python:** Core Programming Language
- **Pandas & NumPy:** Data Manipulation & Numerical Operations
- **Scikit-Learn:** Model Training, Metrics, and Scalers
- **Matplotlib:** Data Visualization & Model Comparison Charts
- **Joblib:** Model Serialization
- **Streamlit:** Interactive Web UI Framework

🚀 Installation & Local Execution

Follow these steps to run the application on your local machine:

1. **Clone the repository:**
   ```
   git clone <span type="placeholder" placeholder-type="person"></span>/house-price-predictor
   ```
2. **Install required dependencies:**
   ```
   pip install -r requirements.txt
   ```
3. **Run the Streamlit application:**
   ```
   streamlit run CODE/house_price_predictor_streamlit_ui.py
   ```

🌐 Deployment

This project is deployment-ready for **Streamlit Community Cloud**:

1. Push your repository files to GitHub.
2. Link your repository on the Streamlit Cloud dashboard.
3. Set the main file path to `CODE/house_price_predictor_streamlit_ui.py` and deploy!

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